Parameter Scope Matters: MNE-L2 for Reliable Low-Latency ANN-to-SNN Conversion

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Abstract

Spiking neural networks (SNNs) support sparse, event-driven computation and can provide energy-efficient inference on neuromorphic hardware. ANN-to-SNN conversion offers a practical route to deep SNNs by reusing well-trained ANN parameters, thereby avoiding the difficulty of training deep SNNs directly. However, the thresholded, discrete representations used by SNNs can be fragile under noise. To investigate how source training shapes this robustness, we study the parameter scope of regularization and identify a failure: applying L2 to batch-normalization (BN) affine parameters can make converted SNNs substantially less robust than applying it only to weights. We analyze this failure through local quantization-boundary crossings and, guided by the analysis, propose margin-normalized effective L2 (MNE-L2). Across five-seed CIFAR experiments, MNE-L2 improves high-noise accuracy over all-parameter L2 by up to 62.06 percentage points while reducing clean firing activity and estimated arithmetic energy. These findings expose regularization scope as an important training choice and demonstrate that MNE-L2 improves the reliability–efficiency trade-off for low-latency SNN inference.

1 Introduction

Neuromorphic processors can exploit sparse, event-driven spikes for efficient inference [1], yet realizing this advantage with deep networks requires accurate representations at short simulation horizons. High-accuracy deep spiking neural networks (SNNs) can be obtained either by direct training or by converting artificial neural networks (ANNs). ANN-to-SNN conversion is particularly practical because it reuses well-trained ANN parameters and avoids the difficulty of optimizing deep SNNs directly [2, 3]. Converted SNNs often require many simulation steps to approximate ANN activations. Reducing the timestep budget lowers latency and event cost but increases finite-time conversion error [4]. Quantized activation training and related calibration methods have substantially improved accuracy at low timesteps [5, 4], making the reliability of these discrete representations under deployment noise increasingly important.

This reliability question is especially important for discrete SNN representations: perturbations can abruptly alter thresholded spike counts. Prior work has studied noise sensitivity in trained SNNs and adversarial robustness after conversion [6, 7]. Most low-latency conversion methods, however, focus on reducing clean conversion error through activation design, calibration, or membrane-potential correction. How the source ANN's training choices shape post-conversion robustness has received much less attention.

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To fill this gap, we start with a controlled ablation of L2 regularization, a standard source-ANN training choice whose parameter scope is often left implicit. In networks with batch normalization (BN) [8], an L2 penalty can cover weights alone or also the BN affine parameters. Figure 1 compares weights-only L2 with penalties that additionally include BN-β, BN-γ, or both. Including BN-γ causes a sharp accuracy collapse even at moderate post-input-IF noise ($\sigma = 1$), whereas weights-only L2 and weights plus BN-β remain stable. This scope-dependent failure motivates our stability analysis.

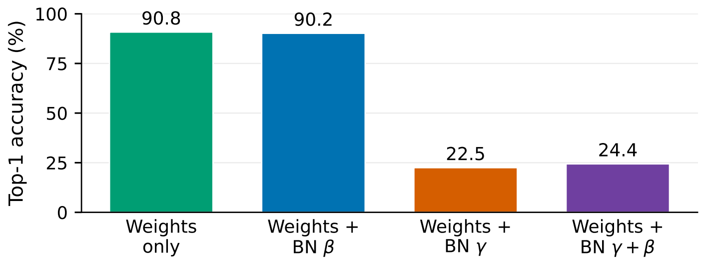

Figure 1: CIFAR-10 VGG16 parameter-scope ablation under post-input-IF Gaussian noise at $L = T = 16$. Accuracy is shown at $\sigma = 1$ for a single seed-42 checkpoint per setting. Including BN-γ, with or without BN-β, causes a sharp collapse.

To explain this scope-dependent failure, we analyze a threshold-quantized converted layer under additive perturbations. A discrete representation changes only when a perturbation crosses a quantization boundary, so local risk is governed by effective noise relative to the nearest margin. Downstream BN-folded weights determine how strongly perturbations are amplified, while IF thresholds determine the conversion margin. Guided by this analysis, we propose margin-normalized effective L2 (MNE-L2), which penalizes BN-folded amplification relative to that margin without directly updating BN affine parameters or IF thresholds through the regularization term.

Our contributions are:

- We identify a previously overlooked L2 parameter-scope failure and isolate BN-γ as the affine term that distinguishes the unstable settings in a controlled four-way ablation.
- We derive a local boundary-crossing view and layerwise margin-to-noise diagnostics that connect effective amplification to changes in discrete SNN representations.
- We formulate MNE-L2 and evaluate it against L1/L2 baselines using five-seed CIFAR VGG13/16/19 experiments, a preliminary ImageNet ResNet-18 extension, and full-test-set firing-density and arithmetic-energy estimates.

2 Related Work

SNN training and ANN-to-SNN conversion. High-performance SNNs are commonly obtained either by direct training or conversion. Direct approaches optimize temporal spiking dynamics through surrogate gradients [9–11], but require backpropagation through multiple simulation steps. Conversion instead trains a source ANN and maps its activations to firing rates. Early methods use weight normalization, threshold balancing, and activation normalization [12, 2]; later work scales conversion to VGG and residual architectures [3]. Our work follows conversion and targets the reliability of intermediate quantized representations at low T, rather than proposing a new neuron model or direct-training rule.

Quantization-based low-latency conversion. Finite-time mismatch has been reduced through potential initialization [13], parameter calibration [14], residual membrane-potential correction [15], offset-spike calibration [16], probabilistic temporal alignment [17], and quantized ANN training [5, 4]. Recent methods also learn conversion-error compensation or bridge quantized activations and spiking neurons [18–20]. These approaches mainly modify conversion dynamics, calibration, or activation design. MNE-L2 is complementary: the conversion rule is unchanged, while the source training objective penalizes layerwise amplification relative to the quantization margin. The BN-scope ablation further shows that the parameter set receiving a nominally standard L2 penalty can materially change measured post-conversion reliability.

Stability and noise sensitivity. Perturbation stability has been analyzed through Lyapunov bounds for ANNs [21], membrane-potential dynamics in trained SNNs [6], spectral properties of random LIF networks [22], and classifier-level adversarial robustness after conversion [7]. We study a different target: whether a controlled post-input-IF perturbation changes intermediate QCFS codes and how these changes propagate through a converted network. The margin-to-noise and bin-crossing diagnostics therefore localize representation failures rather than certifying worst-case classifier robustness.

3 From Parameter Scope to Margin-Aware Regularization

3.1 Low-latency QCFS conversion

For activation x, threshold λ, and quantization level L, the source ANN uses

$$q_L(x;\lambda) = \frac{\lambda}{L}\left\lfloor L\,\mathrm{clip}\Big(\frac{x}{\lambda},0,1\Big) + \frac{1}{2}\right\rfloor. \tag{1}$$

At inference, an integrate-and-fire (IF) layer emits a length-T spike sequence whose rate represents the quantized activation. Unless stated otherwise, we use $L = T = 16$ and a rate-uniform schedule for the VGG experiments. Additional depth-transfer and compact low-latency controls are reported only in the appendix.

3.2 Local noise-to-margin analysis

Let z be the clean output of an IF layer and $\tilde{z} = z+\epsilon$, where $\epsilon \sim \mathcal{N}(0,\sigma^2 I)$. Let $\mathcal{B}_l$ contain the QCFS boundaries of layer l and $d_l(u) = \min_{b\in\mathcal{B}_l}|u-b|$. Under a local additive model $\delta_l \sim \mathcal{N}(0,\sigma^2_{\mathrm{eff},l})$, changing the quantization bin requires crossing a boundary, so

$$\Pr[q_L(u+\delta_l) \neq q_L(u) \mid u] \leq 2\,\mathrm{Q}\bigg(\frac{d_l(u)}{\sigma_{\mathrm{eff},l}}\bigg), \tag{2}$$

where Q is the Gaussian upper-tail function. We estimate $\sigma_{\mathrm{eff},l} = \sqrt{\mathbb{E}[(z_l-\tilde{z}_l)^2]}$, summarize $d_l(u)$ by its median $d_{50,l}$, and report

$$\rho_l = \frac{d_{50,l}}{\sigma_{\mathrm{eff},l}+\varepsilon}, \qquad P_{\mathrm{cross},l} = \Pr[q_L(z_l) \neq q_L(\tilde{z}_l)]. \tag{3}$$

Small ρ_l and large $P_{\mathrm{cross},l}$ identify layers where noise is likely to change discrete spike counts. In an idealized independent aggregation model, σ^2_{eff} is upper-bounded by $O(1/(CT))$, while the nominal QCFS half-cell margin decreases as $1/L$; hence a nominal stability proxy scales as $\sqrt{CT}/L$ (Appendix A). This motivates evaluating precision, width, and timestep jointly rather than treating L as an isolated accuracy knob. It also suggests controlling the BN-folded amplification that contributes to $\sigma_{\mathrm{eff},l}$ relative to the following QCFS margin.

3.3 MNE-L2 derived from the stability analysis

Guided by the preceding analysis, for each convolution or linear layer followed by BN and an IF activation, we fold the BN scale into the matched weight while stopping the regularization gradient through BN and the IF threshold:

$$\widetilde{W}_{l,o} = \mathrm{sg}\bigg(\frac{\gamma_{l,o}}{\sqrt{v_{l,o}+\epsilon_{\mathrm{BN}}}}\bigg)W_{l,o}, \qquad M_{\mathrm{eff},l} = \frac{1}{C_l}\sum_o \|\widetilde{W}_{l,o}\|_F^2. \tag{4}$$

Margin-normalized effective L2 is

$$\mathcal{R}_{\mathrm{MNE}} = \sum_l \frac{L_l^2 M_{\mathrm{eff},l}}{\mathrm{sg}(\lambda_l)^2+\varepsilon}, \tag{5}$$

where sg denotes stop-gradient. Layers without a following IF threshold are excluded. The objective uses the current folded scale to weight each matched W_l, but the penalty does not directly shrink BN-γ, BN-β, or λ_l. This distinction motivates the scope ablation: standard all-parameter L2, weights-only L2, and selective L2 on weights plus BN affine terms are different interventions even when their coefficients are numerically similar.

4 Experiments

4.1 Setup

We use CIFAR-10 and CIFAR-100 [23] with QCFS VGG16 and ImageNet [24] with QCFS ResNet-18 [25]. Unless stated otherwise, $L = T = 16$ and Gaussian noise is injected after the input IF. We denote optimizer L2 applied to all trainable parameters as L2-all. L2-wo restricts the same penalty to convolutional and linear weight tensors, excluding biases, BN affine parameters, and trainable IF thresholds; L1-wo uses the same weights-only scope. The parameter-scope ablation uses CIFAR-10 seed 42 and one stored perturbation stream at each $\sigma \in [0, 5]$. It compares weights-only L2, weights plus BN-β, weights plus BN-γ, and weights plus both affine terms. The cross-dataset CIFAR study evaluates five independently trained checkpoints (seeds 40–44) for L2-all, L2-wo, L1-wo, and MNE-L2. Curves show means and sample standard deviations across training seeds. Appendix C.2 repeats the CIFAR sweep with VGG13 and VGG19 and includes a fixed-α Calibrated MNE control.

The scale-transfer experiment evaluates the same four methods on one ImageNet ResNet-18 checkpoint (seed 42) over $\sigma \in [0, 2]$. Thus, the ImageNet result is a single-checkpoint extension, not a variance-aware benchmark. Hyperparameters are fixed before the noise evaluation, and the reported configurations are not clean-accuracy matched. The injected perturbation is a synthetic internal-noise stress test rather than an adversarial input attack [7] or a substitute for device measurement. CIFAR activity and energy accounting uses all 10,000 test images for each of five seeds at $T \in \{4, 8, 16\}$; only clean activity is used for deployment energy. Layerwise diagnostics use held-out CIFAR-10 examples. Further details are in the appendix.

4.2 BN gain distinguishes the unstable scope settings

Figure 1 isolates the BN affine terms. At $\sigma = 1$, weights-only L2 retains 90.80% accuracy and adding BN-β retains 90.19%. Adding BN-γ instead falls to 22.50%; adding both affine terms reaches 24.41%. At $\sigma = 3$, the first two configurations still achieve 75.63% and 70.65%, while both configurations containing BN-γ are at chance. The near-overlap of the BN-γ and BN-$\gamma + \beta$ curves implicates gain, rather than bias, as the principal scope difference in this controlled run.

The scope result is consistent with the layerwise conversion diagnostics. In late VGG blocks, all-parameter L2 and weights-plus-BN-γ reach $\rho_l \approx 0.05$–0.11 and $P_{\mathrm{cross},l} \approx 0.62$–$0.74$, whereas MNE-L2 and weights-only L2 retain larger margins and mostly stay below 0.15 crossings (Appendix B). This does not establish a universal BN mechanism: the four-way intervention uses one training seed. It shows that reporting "L2" without its parameter scope can conceal experimentally important variation.

4.3 MNE-L2 across CIFAR and ImageNet

Figure 2(a,b) extends the stress test to five independently trained checkpoints per CIFAR dataset. At $\sigma = 3$, MNE-L2 reaches 86.59% on CIFAR-10 and 55.19% on CIFAR-100, compared with 82.01% and 45.33% for weights-only L1. At $\sigma = 5$, the gap widens to 72.06% versus 39.40% on CIFAR-10 and 41.14% versus 17.91% on CIFAR-100. All-parameter L2 collapses near $\sigma = 1$ on both datasets, consistent with the scope ablation.

The high-noise ordering persists for VGG13 and VGG19 across both CIFAR datasets (Appendix C.2), indicating that it is not specific to VGG16 depth.

The single-checkpoint ImageNet result in Figure 2(c) follows the CIFAR ordering at moderate noise: MNE-L2 retains 33.07% at $\sigma = 1$, versus 21.11% for weights-only L1, and remains highest at $\sigma = 2$. This residual, larger-dataset extension is preliminary because it uses one seed.

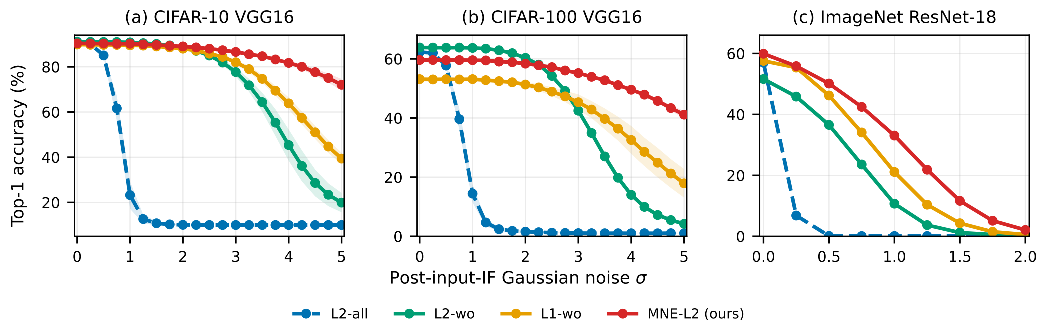

Figure 2: Post-input-IF Gaussian-noise sweeps at $L = T = 16$. (a,b) CIFAR VGG16 curves are five-seed means; bands show sample standard deviations across seeds 40–44. (c) ImageNet ResNet-18 uses one seed-42 checkpoint and therefore has no uncertainty band. MNE-L2 is highlighted in red and all data points use circular markers.

Table 1: Five-seed CIFAR VGG16 robustness–activity summary at $T = 16$ (mean $\pm$ sample standard deviation; 10,000 test images per seed). Accuracy is measured under post-input-IF Gaussian noise at $\sigma = 5$; firing density and arithmetic energy use clean inference. The best value in each dataset and metric is bold, and the second-best is underlined. Clean accuracy and full activity distributions appear in Appendix C.1.

Dataset	Method	Clean firing density	Clean energy (mJ)	Acc. at $\sigma = 5$ (%)
CIFAR-10	L2-all	0.105 ± 0.003	0.585 ± 0.006	10.00 ± 0.00
	L2-wo	0.102 ± 0.001	0.505 ± 0.003	19.92 ± 4.29
	L1-wo	$\mathbf{0.077 \pm 0.001}$	$\mathbf{0.402 \pm 0.005}$	$\underline{39.40 \pm 2.60}$
	MNE-L2 (ours)	$\underline{0.083 \pm 0.001}$	$\underline{0.472 \pm 0.003}$	$\mathbf{72.06 \pm 2.57}$
CIFAR-100	L2-all	0.106 ± 0.003	0.558 ± 0.008	1.05 ± 0.06
	L2-wo	0.105 ± 0.001	0.522 ± 0.002	4.21 ± 0.24
	L1-wo	$\mathbf{0.084 \pm 0.002}$	$\mathbf{0.429 \pm 0.005}$	$\underline{17.91 \pm 4.84}$
	MNE-L2 (ours)	$\underline{0.090 \pm 0.001}$	$\underline{0.494 \pm 0.004}$	$\mathbf{41.14 \pm 1.23}$

4.4 Clean sparsity, robustness, and estimated energy

Table 1 compares all four regularizers at high noise. On CIFAR-10, MNE-L2 lowers clean firing from 0.105 to 0.083 and energy from 0.585 to 0.472 mJ relative to L2-all, while retaining 72.06% rather than 10.00% accuracy at $\sigma = 5$. The same robustness ordering holds on CIFAR-100, where MNE-L2 reaches 41.14% and L2-all reaches 1.05%. L1-wo is the sparsest and least energy-intensive method, but Appendix C.1 shows that it trails MNE-L2 by 6.49 clean CIFAR-100 points.

Following Horowitz [26], we charge repeated first-layer MACs at 4.6 pJ and measured event-triggered ACs in later layers at 0.9 pJ (Appendix Eq. 9). Only clean activity enters energy: CIFAR-10 L2-all firing at $T = 4$ rises from 0.105 to 0.139 under $\sigma = 1$, so noisy activity would confound injected events with deployment cost. Appendix C.1 shows that L2-wo is robust but nearly as active as L2-all; L1 is sparsest, but trails MNE-L2 by 6.49 clean CIFAR-100 points. These estimates omit memory, neuron updates, control, interconnect, and leakage, and are not device measurements.

5 Implications, Limitations, and Conclusion

For deployment, regularizer names should specify their parameter scope: including BN-γ changes the observed robustness ranking. MNE-L2 instead weights convolutional and linear parameters using folded BN statistics without shrinking affine or threshold terms, yielding lower activity than L2-all and an intermediate point between sparse but inaccurate L1 and active, clean-accurate L2-wo. Limitations include a one-seed scope study, one ImageNet checkpoint, synthetic single-site noise, unmatched clean accuracy, and arithmetic-only energy; five-seed CIFAR evidence remains restricted to VGG-family architectures. These constraints preclude universal dominance, but within the tested settings MNE-L2 better preserves post-IF representations than L2-all.

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A A Local Precision–Width–Time Model

Consider an idealized normalized aggregation of independent perturbations,

$$\delta_{C,T} = \frac{1}{CT}\sum_{t=1}^{T}\sum_{i=1}^{C} w_i\epsilon_{i,t}, \qquad \epsilon_{i,t} \overset{\text{iid}}{\sim} \mathcal{N}(0,\sigma^2). \tag{6}$$

If $C^{-1}\sum_i w_i^2 \leq M$, then

$$\mathrm{Var}(\delta_{C,T}) = \frac{\sigma^2}{C^2T}\sum_i w_i^2 \leq \frac{\sigma^2 M}{CT}. \tag{7}$$

For an unclipped QCFS cell, the nominal half-cell margin is $h_L = \lambda/(2L)$. Combining this value with the variance bound gives the nominal proxy

$$\rho_{\text{nom}} = \frac{h_L}{\sigma_{\text{eff}}} \geq \frac{\lambda\sqrt{CT}}{2L\sigma\sqrt{M}}. \tag{8}$$

Table 2: Five-seed clean MNIST conversion at $T = 4$ (mean $\pm$ standard deviation, no injected noise). Δ is L=16 minus L=4 SNN accuracy.

Channels	SNN, $L = 4$	SNN, $L = 16$	Δ
2/4	96.14 ± 0.91	87.80 ± 6.94	-8.34
4/8	98.00 ± 0.26	97.01 ± 0.60	-0.99
8/16	98.67 ± 0.13	98.22 ± 0.37	-0.44
16/32	98.67 ± 0.12	98.42 ± 0.21	-0.25

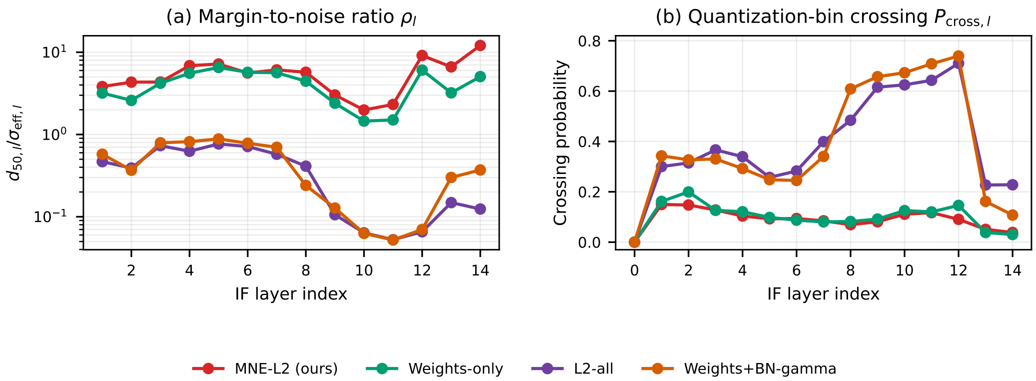

Figure 3: Layerwise CIFAR-10 VGG16 diagnostics at fixed $\sigma = 1$. All-parameter L2 and weights+BN-γ checkpoints develop small margin-to-noise ratios and high bin-crossing probability in later blocks.

Equation (2) follows because a changed quantization code requires $|\delta_l| \geq d_l(u)$ and a zero-mean Gaussian satisfies $\Pr(|\delta_l| \geq d) = 2Q(d/\sigma_{\mathrm{eff},l})$. This local model isolates the competing directions of C, T, and L; it is not a claim that all network perturbations are independent or Gaussian, nor a proof that the empirical accuracy gap must be monotone.

A.1 Compact precision–width control

To separate conversion error from injected-noise calibration, we train weight-decayed MNIST CNNs with channel pairs $(2, 4)$, $(4, 8)$, $(8, 16)$, and $(16, 32)$ at $L \in \{4, 16\}$. Each setting uses five training seeds and is converted at $T = 4$ with no injected noise. Raising L from 4 to 16 improves every source ANN by 0.06–0.29 points, yet reduces $T = 4$ SNN accuracy by 8.34 points for the 2/4-channel model. This loss contracts to 0.25 points at 16/32 channels (Table 2).

B Layerwise Diagnostics

Figure 3 reports the two diagnostics from Eq. (3). Layer 0 is the perturbed input IF; its effective-noise denominator is zero by construction for the downstream-propagation statistic and is omitted from the left panel. The right panel directly measures whether clean and noisy IF outputs differ.

C Additional Experimental Details

The CIFAR VGG16 source ANNs are trained for 300 epochs with batch size 128 and seeds 40–44 where multi-seed results are reported. Training uses SGD with momentum 0.9, initial learning rate 0.1, and cosine annealing. Random crops with four-pixel padding, horizontal flips, CIFAR AutoAugment, and Cutout are used during training; evaluation uses only dataset-specific normalization. MNE-L2 uses coefficient 10^{-4} with detached BN affine/statistical factors and detached IF thresholds. All-parameter L2 uses weight decay 5×10^{-4}, weights-only L1 uses 10^{-5}, and the selective L2 scope controls use coefficient 2.5×10^{-4}. The scope comparison reuses CIFAR-10 seed-42 checkpoints and the same evaluation protocol at every severity.

Table 3: Top-1 accuracy under post-input-IF noise at $\sigma = 1$ (mean $\pm$ sample standard deviation across seeds 40–44).

Dataset	Method	$T = 4$	$T = 8$	$T = 16$
CIFAR-10	L2-all	10.10 ± 0.15	11.17 ± 1.18	23.30 ± 6.46
	MNE-L2 (ours)	88.97 ± 0.21	89.76 ± 0.18	90.09 ± 0.10
CIFAR-100	L2-all	1.56 ± 0.40	2.88 ± 0.83	14.64 ± 3.16
	MNE-L2 (ours)	58.77 ± 0.28	59.27 ± 0.11	59.48 ± 0.23

Table 4: Full clean SNN top-1 accuracy (mean $\pm$ sample standard deviation across seeds 40–44).

Dataset	Method	$T = 4$	$T = 8$	$T = 16$
CIFAR-10	L2-all	90.29 ± 0.23	90.52 ± 0.14	90.59 ± 0.13
	L2-wo	90.57 ± 0.18	90.96 ± 0.06	91.10 ± 0.07
	L1-wo	89.09 ± 0.25	89.70 ± 0.30	89.87 ± 0.20
	MNE-L2 (ours)	89.93 ± 0.16	90.31 ± 0.17	90.38 ± 0.14
CIFAR-100	L2-all	61.83 ± 0.47	62.22 ± 0.49	62.27 ± 0.37
	L2-wo	62.68 ± 0.54	63.57 ± 0.31	63.83 ± 0.26
	L1-wo	52.37 ± 0.95	52.89 ± 0.71	53.13 ± 0.73
	MNE-L2 (ours)	59.24 ± 0.21	59.62 ± 0.31	59.62 ± 0.22

The ImageNet ResNet-18 source ANNs are trained at $T = 0$ for 90 epochs with seed 42, batch size 256, SGD momentum 0.9, initial learning rate 0.1, and cosine annealing. Training applies a random 224×224 resized crop and horizontal flip; validation resizes to 256 pixels and takes a 224×224 center crop. Both splits use standard ImageNet normalization. The highest clean validation-accuracy checkpoint is converted with $L = T = 16$ and evaluated on the 50,000-image validation split. All-parameter and weights-only L2 use coefficient 10^{-4}, MNE-L2 uses 10^{-4}, and weights-only L1 uses 10^{-5}. No coefficient sweep or clean-accuracy matching was performed for the reported ImageNet comparison, and coefficients were not selected from the noisy evaluation curves.

For the CIFAR arithmetic estimate, dense ANN energy and event-driven SNN energy are

$$E_{\mathrm{ANN}} = N_{\mathrm{MAC}} E_{\mathrm{MAC}}, \qquad E_{\mathrm{SNN}} = T N_{\mathrm{MAC}}^{(1)} E_{\mathrm{MAC}} + N_{\mathrm{AC}}^{(>1)} E_{\mathrm{AC}}, \tag{9}$$

with $E_{\mathrm{MAC}} = 4.6$ pJ and $E_{\mathrm{AC}} = 0.9$ pJ. The first convolution receives analog pixels at every timestep; all later convolutional and linear layers use measured presynaptic events. Only clean ($\sigma = 0$) activity is used in this estimate.

C.1 Full CIFAR robustness and energy statistics

Table 3 reports the full $\sigma = 1$ timestep sweep for the main comparison. Tables 4 and 5 report clean accuracy and energy at every timestep; Table 6 adds per-sample activity distributions and the L2-wo/L1 controls at $T = 16$. The main-paper Table 1 reports the four-method $\sigma = 5$ comparison. For energy accounting, we count dense operations in the first convolution at every timestep and event-triggered ACs in all later convolutional and linear layers. Every table uses all 10,000 examples in each test set; uncertainty reflects variation across five independently trained checkpoints, not measurement noise from physical hardware.

Table 5: Full clean CIFAR VGG16 arithmetic-energy estimates (mean ± sample standard deviation across seeds 40–44). ANN energy is 1.5277 mJ on CIFAR-10 and 1.5294 mJ on CIFAR-100. “Ratio” is ANN energy divided by estimated SNN energy.

		$T = 4$		$T = 8$		$T = 16$	
Dataset	Method	Energy (mJ)	Ratio	Energy (mJ)	Ratio	Energy (mJ)	Ratio
CIFAR-10	L2-all	0.1421 ± 0.0015	10.75 ± 0.11	0.2877 ± 0.0030	5.31 ± 0.05	0.5848 ± 0.0060	2.61 ± 0.03
	L2-wo	0.1227 ± 0.0008	12.45 ± 0.08	0.2483 ± 0.0016	6.15 ± 0.04	0.5050 ± 0.0033	3.03 ± 0.02
	L1-wo	0.0975 ± 0.0010	15.67 ± 0.17	0.1969 ± 0.0022	7.76 ± 0.08	0.4019 ± 0.0046	3.80 ± 0.04
	MNE-L2 (ours)	0.1141 ± 0.0005	13.39 ± 0.06	0.2312 ± 0.0012	6.61 ± 0.03	0.4716 ± 0.0026	3.24 ± 0.02
CIFAR-100	L2-all	0.1354 ± 0.0020	11.30 ± 0.16	0.2742 ± 0.0039	5.58 ± 0.08	0.5577 ± 0.0078	2.74 ± 0.04
	L2-wo	0.1271 ± 0.0005	12.03 ± 0.04	0.2569 ± 0.0010	5.95 ± 0.02	0.5225 ± 0.0021	2.93 ± 0.01
	L1-wo	0.1043 ± 0.0011	14.66 ± 0.16	0.2109 ± 0.0025	7.25 ± 0.09	0.4294 ± 0.0052	3.56 ± 0.04
	MNE-L2 (ours)	0.1197 ± 0.0009	12.77 ± 0.10	0.2424 ± 0.0019	6.31 ± 0.05	0.4942 ± 0.0037	3.09 ± 0.02

Table 6: Clean activity distribution, clean arithmetic energy, and $\sigma = 1$ robustness at $T = 16$. Within each seed, firing density is the fraction of nonzero entries across all IF outputs and is summarized over 10,000 samples by its mean, median, and 95th percentile; entries report the mean ± sample standard deviation of these statistics across seeds 40–44. Energy uses clean mean activity only.

Dataset	Method	Clean acc. (%)	Fire mean	Fire median	Fire p95	Energy (mJ)	Acc. at $\sigma = 1$ (%)
CIFAR-10	L2-all	90.59 ± 0.13	0.105 ± 0.003	0.105 ± 0.003	0.120 ± 0.004	0.585 ± 0.006	23.30 ± 6.46
	MNE-L2 (ours)	90.38 ± 0.14	0.083 ± 0.001	0.083 ± 0.001	0.099 ± 0.002	0.472 ± 0.003	90.09 ± 0.10
	L2-wo	91.10 ± 0.07	0.102 ± 0.001	0.103 ± 0.001	0.118 ± 0.001	0.505 ± 0.003	90.82 ± 0.11
	L1-wo	89.87 ± 0.20	0.077 ± 0.001	0.077 ± 0.001	0.090 ± 0.002	0.402 ± 0.005	89.60 ± 0.16
CIFAR-100	L2-all	62.27 ± 0.37	0.106 ± 0.003	0.106 ± 0.003	0.121 ± 0.004	0.558 ± 0.008	14.64 ± 3.16
	MNE-L2 (ours)	59.62 ± 0.22	0.090 ± 0.001	0.090 ± 0.001	0.107 ± 0.002	0.494 ± 0.004	59.48 ± 0.23
	L2-wo	63.83 ± 0.26	0.105 ± 0.001	0.106 ± 0.001	0.120 ± 0.001	0.522 ± 0.002	63.79 ± 0.25
	L1-wo	53.13 ± 0.73	0.084 ± 0.002	0.085 ± 0.002	0.101 ± 0.002	0.429 ± 0.005	53.08 ± 0.76

C.2 Depth transfer across VGG13 and VGG19

We repeat the post-input-IF Gaussian-noise sweep at $L = T = 16$ using VGG13 and VGG19 on CIFAR-10 and CIFAR-100. Each curve reports the mean and sample standard deviation across independently trained seeds 40–44. In addition to the four main regularizers, this control includes Calibrated MNE with the fixed $\alpha = 0.1$ configuration; α is a prescribed training schedule rather than a gradient-triggered or adaptive choice.

Across VGG13/16/19, MNE-L2 attains $74.28/72.06/73.23\%$ at $\sigma = 5$ on CIFAR-10 and $44.01/41.14/37.60\%$ on CIFAR-100. Calibrated MNE closely follows L2-wo despite matching or slightly exceeding its clean accuracy, while L1-wo incurs a substantial CIFAR-100 clean-accuracy cost that grows from VGG13 to VGG19. Thus the high-noise MNE-L2 advantage transfers across the three completed depths, although the CIFAR-100 endpoint declines with depth.

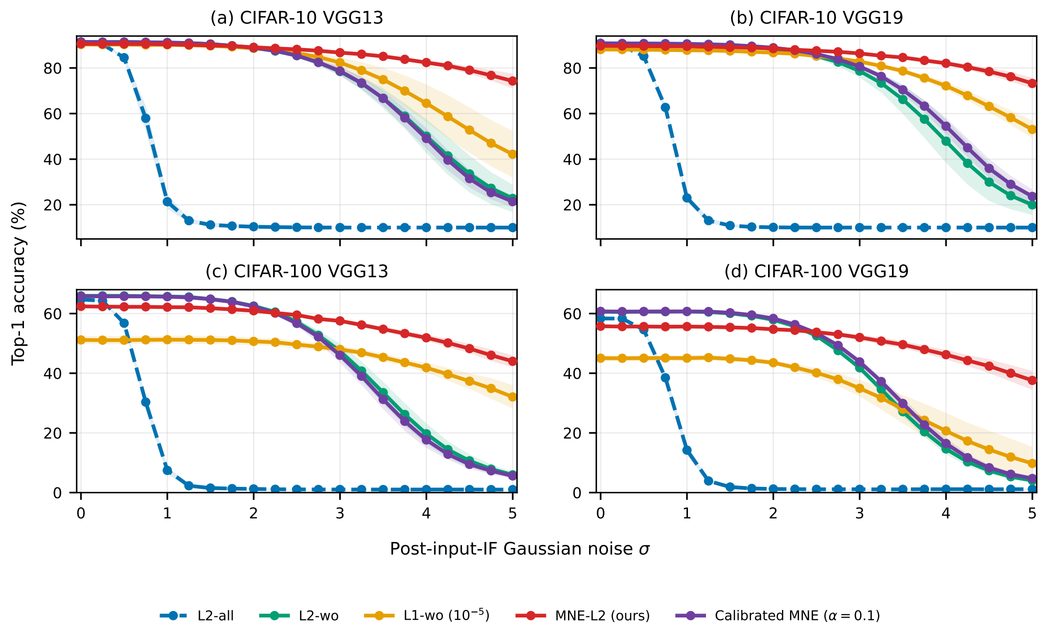

Figure 4: Depth-transfer robustness at $L = T = 16$. Gaussian noise is injected after the input IF. Curves are five-seed means and bands show sample standard deviations. MNE-L2 is highlighted in red; Calibrated MNE is included only as an appendix control.

Table 7: Additional-depth accuracy endpoints (mean $\pm$ sample standard deviation across seeds 40–44). The best value within each architecture, dataset, and noise condition is bold.

Architecture	Method	CIFAR-10 $\sigma = 0$	CIFAR-10 $\sigma = 5$	CIFAR-100 $\sigma = 0$	CIFAR-100 $\sigma = 5$
VGG13	L2-all	90.93 ± 0.14	9.99 ± 0.01	64.73 ± 0.19	1.01 ± 0.01
	L2-wo	$\mathbf{91.44 \pm 0.14}$	22.78 ± 5.99	$\mathbf{65.94 \pm 0.15}$	5.96 ± 0.96
	L1-wo	90.30 ± 0.14	42.17 ± 10.20	51.16 ± 1.31	32.06 ± 3.92
	MNE-L2 (ours)	90.61 ± 0.19	$\mathbf{74.28 \pm 3.25}$	62.41 ± 0.15	$\mathbf{44.01 \pm 2.02}$
	Cal. MNE	91.41 ± 0.13	21.38 ± 2.58	65.82 ± 0.22	5.59 ± 0.92
VGG19	L2-all	90.19 ± 0.12	10.00 ± 0.00	58.37 ± 0.64	1.13 ± 0.18
	L2-wo	90.69 ± 0.18	19.95 ± 4.44	60.62 ± 0.32	4.03 ± 0.65
	L1-wo	88.09 ± 0.25	53.04 ± 3.85	45.04 ± 1.01	9.77 ± 5.59
	MNE-L2 (ours)	89.67 ± 0.06	$\mathbf{73.23 \pm 2.50}$	55.75 ± 0.43	$\mathbf{37.60 \pm 3.19}$
	Cal. MNE	$\mathbf{90.84 \pm 0.03}$	23.66 ± 3.05	$\mathbf{60.73 \pm 0.22}$	4.73 ± 0.76

C.3 Compact MNIST-family controls

We use MNIST [27] and Fashion-MNIST [28] with a 2,081-parameter CNN: two 3×3 convolution-BN-IF blocks with 2 and 4 channels, 2×2 max pooling, and a linear classifier. Source ANNs are trained for 30 epochs with seeds 40–44 and converted at $T \in \{4, 8, 16\}$. We compare MNE-L2, orthogonal weight regularization, no regularization, and optimizer L2. During the low-latency sweep, Gaussian noise is added after both the input IF and first hidden IF at $\sigma \in \{0, 0.25, 0.5\}$; each nonzero point averages three paired streams within each training seed.

For the compact CNN, the dense ANN baseline is 30,184 MACs per sample. The event counter charges every nonzero presynaptic activation once for each valid downstream connection. It counts both convolutional layers and the classifier, including edge effects from padding, and accumulates over all T timesteps. Reported means and standard deviations are across five independently trained checkpoints.

Figure 5 gives the compact low-latency view. At $T = 4$, the event-SynOps ratio ranges from 0.36 to 0.45 on Fashion-MNIST and 0.63 to 0.65 on MNIST; at $T = 16$, it rises to 1.54–1.97 and 2.71–2.86. These are algorithmic operation counts rather than wall-clock latency or hardware energy [1, 26].

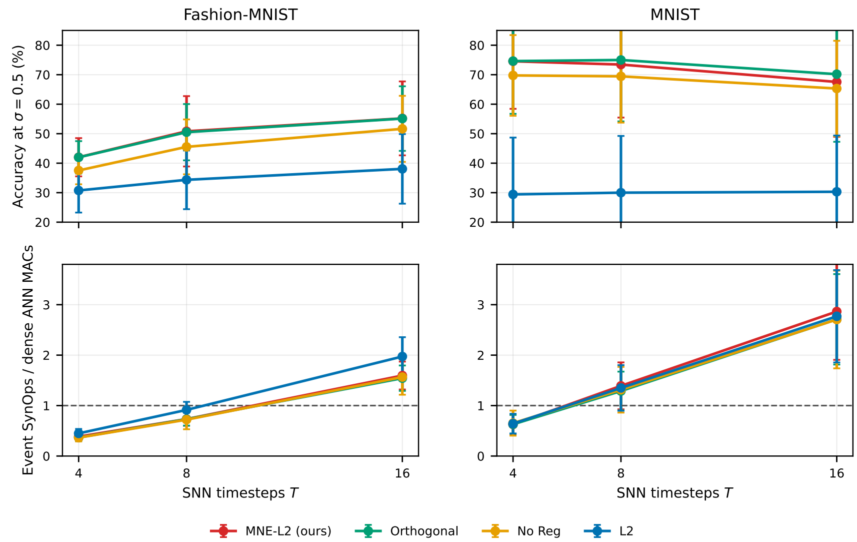

Figure 5: Appendix compact-model control. Top: five-seed accuracy under two-site Gaussian noise with $\sigma = 0.5$. Bottom: event-driven synaptic-operation proxy relative to one dense ANN MAC pass; the dashed line marks ratio 1. Error bars are standard deviations across training seeds.

Table 8: Full five-seed accuracy at $\sigma = 0.5$ (mean $\pm$ standard deviation across training seeds). Noise is injected after both the input IF and first hidden IF.

Dataset	Method	$T = 4$	$T = 8$	$T = 16$
Fashion	MNE-L2 (ours)	42.01 ± 6.48	50.81 ± 11.93	55.19 ± 12.53
Fashion	Orthogonal	41.98 ± 5.49	50.50 ± 9.54	55.12 ± 10.95
Fashion	No Reg	37.53 ± 4.69	45.53 ± 9.30	51.64 ± 11.19
Fashion	L2	30.75 ± 7.49	34.35 ± 9.95	38.06 ± 11.80
MNIST	MNE-L2 (ours)	74.55 ± 16.12	73.44 ± 17.98	67.54 ± 18.60
MNIST	Orthogonal	74.62 ± 17.89	74.99 ± 20.82	70.17 ± 22.92
MNIST	No Reg	69.75 ± 13.68	69.44 ± 15.69	65.31 ± 16.18
MNIST	L2	29.41 ± 19.27	30.00 ± 19.23	30.30 ± 19.07

D Reproducibility Notes

All reported datasets use their standard public train/test splits. The evaluation code fixes the training seed separately from the perturbation seed, stores every per-seed result before aggregation, and reports sample standard deviation rather than standard error. Figures 1, 2, and 4 are generated directly from stored CSV summaries by a deterministic plotting script; curve values are neither redrawn nor transcribed. The compact event-count control uses CPU inference and 1,024 examples per checkpoint. The CIFAR activity evaluation uses each full 10,000-example test set: sample mean, median, and p95 are computed within each seed, then summarized across seeds 40–44 by mean and sample standard deviation. Only $\sigma = 0$ activity enters energy; $\sigma = 1$ activity is retained as a diagnostic and excluded from deployment estimates. Full VGG and ResNet training requires GPU compute; evaluation does not update model parameters.

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